

Beyond Seen Mistakes: A Matched Test of Natural Error-Composition Generalization in Multi-Criterion Skill Assessment

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Abstract—Automated skill assessment—of exercise form, of clinical technique—predicts several criteria at once, yet grouped evaluation does not isolate whether a familiar diagnosis was present in training. We introduce a recording-pair-disjoint matched test on public ALEX-GYM-1 poses. For each of 12 natural diagnoses, target-present and target-absent models use identical test recordings, validation data, initialization, class weights, and training-set size. Across four temporal backbones—TCN, BiGRU, Transformer, and state-space—an optimized exchange minimizes residual criterion-marginal differences. Removing target-diagnosis examples then lowers exact diagnosis match from 22.65% to 1.52% (target bootstrap gap 21.13 points, 95% CI [9.86, 34.73]; two-sided $p = .00049$); bit accuracy falls 12.80 points. Every target falls, and the largest gaps occur where residual imbalance is smallest. We then replicate the whole protocol on CPR-Coach, which annotates 13 error types and releases naturally co-occurring composites: across 72 eligible compositions the gap is 11.80 points, 95% CI [9.03, 14.77], inside the ALEX-GYM-1 interval but six times more precisely bounded, and the primary test is no longer pinned to its own floor. A label-space audit finds that one-bit opposing neighbors predict vulnerable criteria across models, yet add nothing once ordinary criterion-state support is controlled for ($p = .200$ on ALEX-GYM-1; $p = .87$ over 936 units on CPR-Coach). Finally, the second dataset exposes a representation defect the first conceals: resampling every repetition to a fixed frame count erases the quantity that rate-defined criteria measure, leaving four CPR criteria 14.50 points below their own trivial baselines. Restoring duration, speed and cadence recovers them and raises exact match from 15.72% to 37.47%. The contribution is a controlled composition test, replicated externally, and an evidence-bounded diagnosis of its failure.

Index Terms—skill assessment, exercise assessment, multi-label classification, compositional generalization, skeleton sequences, label-space diagnostics

I. INTRODUCTION

Fine-grained exercise assessment differs from ordinary action recognition. A system must report several simultaneous errors—for example foot orientation, hip position, back posture, and movement depth—rather than one action class. Consequently, a label is a binary diagnosis vector $\mathbf{y} \in \{0,1\}^C$, and the practically relevant question is whether individually familiar criterion states remain recognizable when they occur in a new combination.

The distinction is easy to miss. A recording-disjoint random split protects against source overlap but does not deliberately hold out a chosen diagnosis vector, so a model can obtain reasonable marginal accuracy while using pose patterns

correlated with entire diagnoses. This matters for feedback: one systematically wrong criterion invalidates an otherwise plausible diagnosis.

Composite errors have been studied directly by CPR-Coach, which trains on isolated error classes and tests synthetically enumerated composites [1]. Held-out label combinations are also established in multi-label learning [2]. Our setting is narrower and naturalistic: training remains multi-error, every evaluated criterion state has training support, and test diagnoses are naturally observed in a public two-view exercise dataset, so we do not claim the first composite-error task or combination split. The difference is the control: a conventional composition split asks whether unseen combinations are harder, whereas we hold the evaluation recordings fixed and ask whether they become harder when that one diagnosis is removed from otherwise supported training.

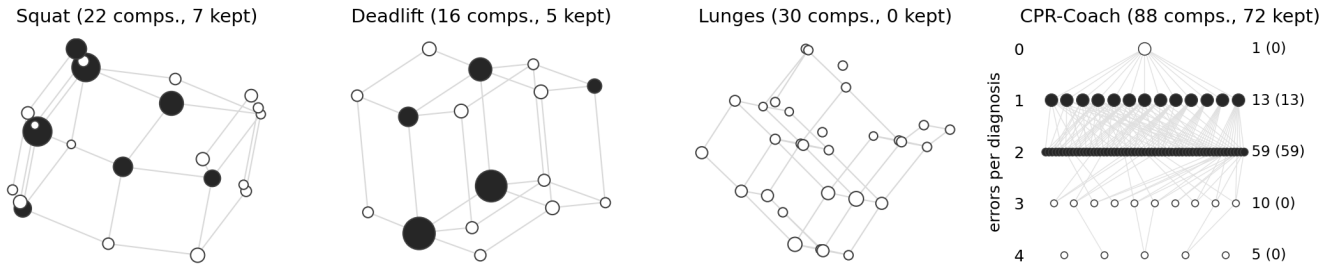
We organize the study around two retrospective questions. **RQ1** asks whether removing a target diagnosis from training reduces performance on the same held-out recording groups under matched training size and loss weights. **RQ2** asks whether local one-bit label structure identifies vulnerable criteria beyond ordinary marginal criterion support. Tests are two-sided; exact match was fixed as the sole primary endpoint before the corrected matched rerun.

We contribute (i) a data-audited observed-composition protocol, (ii) a controlled target-present/absent comparison, (iii) an external replication of both research questions on a second, independently collected composite-error dataset, (iv) evidence that a plausible local label diagnostic is subsumed by ordinary per-criterion class imbalance, and (v) the finding that fixed-length temporal resampling, a near-universal preprocessing step, silently removes the evidence for rate-defined criteria. Figure 1 shows the naturally observed diagnosis spaces and targets retained by the support rule.

II. RELATED WORK

A. Exercise assessment and composite errors

ALEX-GYM-1 provides two-view 3D poses and multi-criterion exercise annotations [3]. Other rehabilitation datasets emphasize clinical scores, correctness, or multi-task assessment rather than natural held-out multi-error label sets [4]–[6]. FineDiving and FineParser study procedure-aware action quality at fine temporal granularity [7], [8], but not this diagnosis-set



Node area = diagnosis frequency; filled nodes are retained by the frozen support rule; edges join diagnoses differing in one criterion.

Fig. 1. Observed diagnosis spaces of both datasets, with the count each retains under the frozen support rule. ALEX-GYM-1’s exercises project legibly under PCA; CPR-Coach’s 88 compositions over 13 criteria do not, so they are drawn by cardinality, which is how their structure actually runs. Lunge retains no target because its candidates cannot preserve the required train/validation state support after recording-group removal, and CPR-Coach’s triples and quadruples fail the same rule.

TABLE I
BOTH PROCESSED DATASETS. “GROUPS” ARE THE SPLIT UNIT: PAIRED RECORDING IDENTIFIERS FOR ALEX-GYM-1 AND COMPOSITION-PURE CLASS FOLDERS FOR CPR-COACH. NEITHER IS A VERIFIED PARTICIPANT IDENTITY.

Dataset	Task	Reps.	Crit.	Diag.	Groups
ALEX-GYM-1	Squat	294	6	22	59
	Deadlift	269	5	16	55
	Lunge	106	7	30	53
CPR-Coach	Compression	1,414	13	88	234

shift. CPR-Coach is the closest direct composition formulation [1]; its isolated-error training regime differs from our naturally multi-error training pool.

Motion-focused tokenization reduces background-driven video domain shift [9], an adjacent evidence-selection principle rather than a baseline for pose input with no background.

B. Multi-label shortcuts and unseen combinations

Multi-label models often exploit label dependence. Compositional multi-label classification explicitly holds out label combinations [2], while counterfactual and correlation-shortcut work studies harmful co-occurrence dependence [10], [11]. We use this literature to motivate a diagnostic, not a causal claim: label counts characterize exposure in the benchmark and never enter the exercise classifier. Opposing-state support is target-relative criterion-state imbalance, an ordinary quantity whose link to degraded minority-state generalization is established [12]–[14]; our contribution is showing it subsumes a more localized label-neighborhood account under composition shift.

III. DATA AND PROTOCOL

A. Dataset and preprocessing audit

The public ALEX-GYM-1 subset contains squat, Romanian deadlift, and lunge repetitions with frontal and lateral 33-joint 3D pose sequences. We convert each trainer criterion to an error bit: one denotes a violated criterion. Table I summarizes the processed data.

All-zero frames occur in both views, ranging from 1.95% (lunge frontal) to 14.87% (squat lateral); per-exercise, per-view

counts are released. We linearly interpolate whole missing frames before pelvis centering, robust body-scale normalization, and resampling to 16 frames. One squat repetition has no valid frontal frame and is excluded. Raw-file SHA-256 hashes, the exclusion, and per-sequence quality counts are released. Isolated zero joints are not imputed, since their semantics cannot be verified.

B. Observed-composition holdout

For exercise e , let s_i identify a paired frontal/lateral recording and y_i its diagnosis. For target diagnosis y^* , the test set is all repetitions with $y_i = y^*$. Every recording group containing that target is removed from the train/validation pool, including its non-target repetitions. The remaining groups are split deterministically. Candidate splits are rejected unless both states of every criterion occur at least eight times in training and at least once in validation. Train-validation, train-test, and validation-test group intersections are zero in every fold.

A target is eligible when it has at least ten repetitions, three source groups, and a feasible supported split. Seven squat and five deadlift diagnoses qualify. Two lunge candidates are infeasible under that rule—their minority pool holds exactly eight examples—and are excluded rather than evaluated under a weaker one. The primary experiment is therefore 12 targets $\times$ three seeds per model.

Ordinary validation is a secondary comparator, not a pure seen-composition control: 409 of 1,760 validation rows (23.2%) have a diagnosis absent from that fold’s training subset.

C. Matched composition-presence test

RQ1 instead holds the evaluation data fixed. For each target and seed, one target recording group is test-only; the three seeds select distinct groups. Validation uses groups containing no target repetition. The *present* training set contains target repetitions from the remaining target groups. The *absent* set exchanges those rows for an equal number of non-target rows from the same training-group pool. Because every eligible replacement differs from the target in at least one bit, exact criterion-marginal equality is impossible. Before training, a deterministic binary optimization therefore minimizes the

largest per-criterion positive-count difference and then their sum. Relative to random exchange, their fold-mean maximum falls from 22.06 to 10.89 counts and their total from 55.08 to 33.42. Thus the conditions have identical size, validation and test rows, initialization, shared class weights, and zero group overlap, with reduced but nonzero marginal differences. This test uses four generic temporal backbones and averages seeds, then models, within each diagnosis.

D. External replication on CPR-Coach

ALEX-GYM-1 alone cannot show that the effect is not a property of one dataset, so we repeat the entire protocol on CPR-Coach [1], which annotates 13 error types and releases recordings in which two, three or four co-occur. Only its released 2D pose keypoints and labels are used. Each take is recorded on four synchronised channels, which are concatenated into one repetition rather than treated as four rows. Poses are Halpe body joints; the 42 hand keypoints are discarded because their mean detector confidence is .058 against .606 for the body, so three hand-placement criteria are effectively unobservable from the released skeletons. The recording group is the composition-pure class folder, of which every eligible composition has exactly three, matching the three-seed distinct-test-group rule.

Applying the same frozen support rule yields 72 eligible compositions—13 single-error and 59 double-error—from 1,414 repetitions, against 12 on ALEX-GYM-1 (Fig. 1). The larger pool also balances better: the optimized exchange cuts the fold-mean worst-criterion mismatch from 10.78 to 3.71 counts, roughly three times smaller than is achievable on ALEX-GYM-1. Labels are sparser here (1.67 positives of 13), so an all-zero predictor already scores 87.17% bit accuracy against 2.97% exact match; exact match therefore remains the primary endpoint and bit accuracy is read against that baseline throughout.

E. Models and training

We use six compact controlled models, all receiving the same 16×198 sequence (two views $\times$ 33 joints $\times$ xyz). The four matched backbones are a dilated TCN, a bidirectional GRU, a two-layer four-head Transformer encoder, and a bidirectional selective scan, with 112k, 99k, 190k and 13k deadlift parameters; layer dimensions are in the released code. The SSM is a compact study implementation, not an official reproduction. ST-GCN applies two spatial graph/temporal convolution blocks to each MediaPipe view using the physical skeleton edges, then fuses pooled view features (35k deadlift parameters).

FACT routes each criterion to the view and joint set its annotation wording names, through a learned residual on a sigmoid mask, then pools per-view temporal evidence into one 32-D token per criterion, decoded by a local linear score plus a learned two-state prototype distance. It never consumes other labels at inference and has 84,858 squat and 81,048 deadlift parameters. It is a locality-oriented comparator, not a claimed

superior architecture; exact layer dimensions are in the released code.

Every model receives the same representation and split, with class-weighted binary cross-entropy, AdamW, fixed 0.5 thresholds, and early stopping on validation macro-F1. Hyperparameters are never selected using the held-out target. We report target-equal means: seeds are averaged within target, then all 12 targets receive equal weight. The full-target and matched experiments use separate deterministic split constructions; each released manifest records raw-file hashes, preprocessing version, exact indices, target support, exclusions, and zero group intersections for every target and seed.

IV. LABEL-SPACE DIAGNOSTIC

For criterion c , let $\mathbf{y}_{-c}^*$ denote the other criterion bits. Label-Opposition Pressure is

$$LOP_c(\mathbf{y}^*) = \sum_{i \in \mathcal{T}} \mathbb{I}[\mathbf{y}_{i,-c} = \mathbf{y}_{-c}^*] \mathbb{I}[y_{i,c} \neq y_c^*], \quad (1)$$

where $\mathcal{T}$ is the fold training set. Large pressure means many training examples reproduce the target context while supporting the wrong current bit. We compare it with global opposing-state support, $G_c = \sum_{i \in \mathcal{T}} \mathbb{I}[y_{i,c} \neq y_c^*]$, a single training-label count requiring no context matching. Target frequency, cardinality, and total one-bit neighborhood density are constants across criteria inside a diagnosis and cancel under within-diagnosis ranking. Both LOP and G_c are exposure associations, not causal influence or a guarantee of failure.

V. STATISTICAL ANALYSIS

The held-out diagnosis is the statistical unit. For RQ1, we average seeds and the four matched architectures within target, then test 12 present-absent differences. Exact match is primary; bit accuracy and F1 are secondary. For RQ2, we average all five original temporal models over seeds, compute Spearman correlation within diagnosis, and take the raw diagnosis-equal mean across 67 target-criterion units. A 50,000-shuffle test permutes a predictor only among criteria of the same diagnosis. A partial rank test residualizes LOP and accuracy against G_c and target-bit state before applying the same blocked permutation. We report 100,000-sample target bootstrap intervals and two-sided probabilities. At 12 targets the exact two-sided Wilcoxon minimum is $2/2^{12} = .00049$; a p at that value means all targets moved alike and cannot fall further. Exact match was fixed before both corrected reruns; the exchange optimization was specified before examining its model outcomes.

VI. RESULTS

A. RQ1: matched removal exposes diagnosis dependence

Under optimized exchange and matched test recordings, target-present exact match is 22.65%, versus 1.52% when the target is absent from training: a 21.13-point gap (95% CI [9.86, 34.73]; two-sided Wilcoxon $p = .00049$). The gap is positive for every architecture (26.76 TCN, 20.69 BiGRU, 20.14 Transformer, and 16.94 SSM points). Bit accuracy also

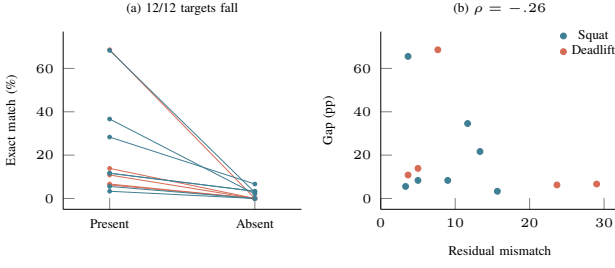


Fig. 2. (a) Exact diagnosis match for each of the 12 held-out diagnoses, under matched training size, class weights, initialization, validation and test rows. (b) The same gaps against the residual worst-criterion mismatch the exchange optimization could not remove. Color marks exercise.

TABLE II
TARGET-EQUAL ALEX-GYM-1 PERFORMANCE (%) IN THE SECONDARY FULL-TARGET STRESS TEST. VALIDATION IS ORDINARY GROUP-HELD-OUT; LOCO DELIBERATELY WITHHOLDS ONE NATURAL DIAGNOSIS.

Model	Validation			LOCO		
	Bit	μ F1	Exact	Bit	μ F1	Exact
TCN	65.08	56.35	12.61	63.04	44.86	2.39
BiGRU	65.32	57.00	14.71	60.61	42.11	2.63
Transformer	64.26	57.08	12.20	59.83	41.79	2.07
Selective SSM	66.89	56.91	14.23	62.50	41.82	3.41
ST-GCN	64.61	56.08	10.86	65.87	49.82	2.85
FACT	72.39	63.95	18.56	64.89	44.77	5.14

falls 12.80 points (CI [7.38, 18.79]; $p = .00049$). The original random-exchange analysis gives a compatible 25.44-point exact-match gap (CI [17.31, 33.32]). RQ1 is therefore supported without comparing different evaluation distributions and is robust to substantially improved marginal balance.

Figure 2 resolves the 12 targets individually. All 12 fall, and residual mismatch does not track the effect: the two largest gaps (68.61, 65.56 points) sit at two of the smallest residual mismatches (7.7, 3.7 counts), and the gap–mismatch rank association is negative ($\rho = -.26$), opposite the confound’s prediction. This is evidence against residual marginal mismatch being the main explanation, not a demonstration that no confounding remains; at 12 targets it is suggestive, not decisive.

The secondary full-target stress test still measures a harder distributional setting. In Table II FACT leads every validation column and LOCO exact match, while ST-GCN leads LOCO bit accuracy and micro-F1. FACT’s LOCO advantage over TCN is unresolved: +1.85 bit points ($p = .204$) and +2.75 exact points ($p = .086$) in two-sided exploratory comparisons. We therefore make no architecture-superiority claim. Micro-F1 is pooled across criterion decisions; macro-F1 is omitted because every criterion has one constant state inside a target fold, making it meaningless here. “Bit” is the fraction of all criterion decisions that are correct; “Exact” requires every criterion of a repetition to be correct—so FACT’s 64.89% LOCO Bit but 5.14% Exact means most individual decisions are right while few complete diagnoses are.

B. RQ1 replicates on a second dataset

Table III repeats the matched test on CPR-Coach. The headline comparison deliberately uses the unmodified representation, so that it reproduces the pre-specified ALEX-GYM-1 protocol

TABLE III
THE SAME PROTOCOL ON BOTH DATASETS. CPR-COACH NUMBERS ARE THE FOUR MATCHED BACKBONES, AS ON ALEX-GYM-1.

	ALEX-GYM-1	CPR-Coach
Eligible compositions	12	72
Exact match, target present (%)	22.65	15.72
Exact match, target absent (%)	1.52	3.92
Gap (points)	21.13	11.80
95% CI (points)	[9.86, 34.73]	[9.03, 14.77]
Compositions falling	12/12	62/72
RQ2 partial LOP, controlled	-.182	-.005
RQ2 two-sided p	.200	.87

rather than a better one; the improved condition is reported separately in Sec. VI-D. Removing the target composition lowers exact match from 15.72% to 3.92%, a gap of 11.80 points (95% CI [9.03, 14.77]), with 62 of 72 compositions falling, 8 tied and 2 reversed. The gap is positive for every backbone (12.57 Transformer, 12.18 BiGRU, 11.78 TCN, 10.67 SSM).

The residual-imbalance argument takes a different form here. The larger pool lets the exchange balance so well that the residual is both smaller and nearly uniform across targets (fold-mean worst-criterion mismatch 3.71 counts), so it has too little between-target variation to explain the gaps at all; its rank association with the gap is +.044 ($p = .71$). On ALEX-GYM-1, where the residual varies substantially, the same argument had to be made through the correlation.

Two things matter more than the point estimate. The CPR-Coach estimate lies inside the ALEX-GYM-1 interval while being roughly six times narrower, so the two datasets are statistically compatible rather than merely same-signed. And with 72 units the primary test is no longer saturated: on ALEX-GYM-1 the exact Wilcoxon p of .00049 is the smallest value attainable at $n = 12$, whereas here the exact value is 7.4×10^{-18} against a floor of 5.4×10^{-20} , a factor of 137 above it.

Two architectures added on this dataset do not support the effect. ST-GCN reaches only 2.22% exact match against 0.35%, with 64 of 72 compositions tied at zero, so it fails the task outright rather than failing it compositionally. FACT, given the frontal and lateral channels identified from measured geometry, shows no gap under its anatomical prior (−3.40 points) and a smaller one under a random joint map of equal size (+4.86); we therefore report no transfer of its anatomical prior to this dataset.

C. RQ2: marginal support, not local opposition, is the practical predictor

Raw LOP is negatively associated with held-out criterion accuracy across all five temporal models (per-model ρ from −.367 to −.648, all blocked $p \leq .00584$); the model-mean ρ is −.632 ($p < .00004$). However, global opposing-state support is stronger ($\rho = -.742$, $p < .00004$). After controlling for that support and the target bit, LOP’s partial rank association is −.182 ($p = .200$). RQ2 is therefore not supported in its

incremental form: the local count is descriptive, but these data favor the simpler marginal-support account.

The association is practically large: among the 67 target-criterion units, the bottom G_c quartile (least opposed) averages 82.70% held-out accuracy versus 40.79% for the top quartile (most opposed). G_c is an ordinary, target-relative imbalance count over training labels, so this confirms—rather than introduces—a training-free audit signal for a proposed composition, available without running an experiment.

The negative result replicates, and more decisively. Over 936 target-criterion units on CPR-Coach, LOP’s partial rank association after the same controls is $-.005$ ($p = .87$); the pre-specified sensitivity that drops the three unobservable hand criteria gives $+.020$ ($p = .605$) over 720 units. Raw associations are consistent across all five architectures there ($-.231$ to $-.316$, each $p = 10^{-4}$). One claim does not carry over and we do not restate it: on ALEX-GYM-1 marginal support was the stronger raw predictor ($\rho = -.742$ against $-.632$), whereas on CPR-Coach that ordering reverses ($-.255$ against $-.316$). What survives on both datasets is the null—local opposition adds nothing once ordinary support is controlled—not the ranking of the two raw predictors.

D. Fixed-length resampling erases rate-defined criteria

Resampling every repetition to 16 frames normalizes the time axis, which removes the quantity that criteria such as “Slow Frequency” or “Insufficient Pressing” are defined by: a two-second and a six-second repetition become the same tensor. Measured against each criterion’s own trivial baseline on CPR-Coach, the four rate-defined criteria score 14.50 points *below* it while the six postural criteria score 7.71 *above*—the models do worse than guessing exactly where the evidence has been deleted.

Restoring three statistics computed before resampling—log duration, mean displacement per original frame, and the dominant non-DC frequency of the highest-variance coordinate—recovers them. Rate criteria gain 17.73 points of lift while postural criteria move 0.67 and hand criteria -0.56 , so the channels supply rate information rather than label information. Exact match rises from 15.72% to 37.47% and the composition gap from 11.80 to 27.15 points, with 70 of 72 compositions falling; bit accuracy rises from 86.27%, below the all-zero baseline, to 91.90% above it. On ALEX-GYM-1 the same channels change nothing (gap 21.13 to 19.05 points, rate-worded criteria -2.12), which is consistent: squat and deadlift criteria are largely geometric, whereas chest compression is intrinsically a rate task. The defect is therefore severe for rate-defined criteria and immaterial for postural ones—a property of the annotation vocabulary, not of either dataset.

VII. DISCUSSION

The matched comparison changes the evidential strength: target presence is varied while evaluation recordings and training budget are fixed. The effect appears in convolutional, recurrent, attention, and state-space models, and ST-GCN’s

low full-target exact match shows explicit skeleton topology is not by itself a remedy.

The control analysis also changes the mechanism claim. LOP reliably describes which bit fails, but it is largely a proxy for target-relative criterion-state imbalance (Sec. II-B). The practical takeaway is therefore not a label-neighborhood mechanism but that this ordinary quantity, on its own, predicts held-out failure well enough to serve as a deployment-time check requiring no training run.

The second dataset does more than widen coverage. It moves the primary test off its own floor, turns a same-signed result into a statistically compatible one, and—because its annotation vocabulary is rate-heavy where ALEX-GYM-1’s is geometric—exposes a preprocessing defect that a single dataset of either kind would have hidden. That the same three channels are decisive on one dataset and inert on the other is itself the evidence that the defect tracks what a criterion measures rather than which corpus it came from.

VIII. LIMITATIONS

Two datasets is still not many, and both remain recording-disjoint rather than participant-disjoint: neither release carries verified participant identities, so repeated performers may span folds. Three CPR-Coach criteria are effectively unobservable from its released skeletons, and its compositions are thinner (mostly pairs) than ALEX-GYM-1’s. FACT and ST-GCN results on CPR-Coach come from a reduced protocol—one seed for FACT—and its anatomical joint map, like ALEX-GYM-1’s, is a manual reading of the domain, which is why FACT stays a comparator and why the random-map control is reported beside it. The restored rate channels are also only a partial remedy: they recover the frequency- and depth-defined criteria but leave Jump Pressing and Excessive Pressing still below their trivial baselines, so duration, speed and cadence do not capture everything those criteria depend on. The matched test uses three test groups per diagnosis holding 1–5 repetitions (median 5), making per-fold exact match coarse, and exchanges examples rather than preserving identical marginals; optimization roughly halves the mean worst-criterion difference, but residual imbalance prevents a causal interpretation. Thresholds are fixed at 0.5, with no calibrated-probability claim. Missing-frame rates reach 14.87% in the squat lateral view; interpolation reduces but cannot eliminate pose-detector uncertainty. The G_c quartile split (17 units/quartile) reuses the RQ2 sample, so it is descriptive, not independent validation. Exact match is clinically intuitive but harsh; we also report bit accuracy and micro-F1.

IX. CONCLUSION

On identical held-out recordings, removing a natural diagnosis from matched training reduces exact match by 21.13 points on ALEX-GYM-1 and 11.80 points across 72 compositions of CPR-Coach, the two intervals overlapping and the second no longer pinned to the smallest value its test can return. Label-space analysis predicts which criteria fail, but local one-bit opposition adds nothing beyond ordinary criterion-state support

on either dataset. The second corpus also shows that resampling repetitions to a fixed frame count erases the evidence for rate-defined criteria, pushing them below their own trivial baselines until duration, speed and cadence are restored. The result is a reproducible composition test, externally replicated, an evidence-bounded diagnosis of its failure, and a preprocessing caution for any multi-criterion assessment whose vocabulary names a rate.

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